

GIUSEPPE PEDONE

Software Engineer

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SUMMARY

Research Engineer with a strong background in computer engineering and applied research on intelligent systems, distributed architectures, and embedded AI. Experienced in designing and validating AI-driven solutions for industrial, real-time, and cyber-physical systems, with a focus on anomaly detection, predictive maintenance, and system-level optimization. Strong interest in the integration of artificial intelligence into operating systems and low-level software layers to enhance performance, adaptability, and autonomy of complex systems.

RESEARCH INTERESTS

- Machine Learning for Cyber-Physical and Industrial Systems
- Distributed and Real-Time AI Architectures
- Anomaly Detection and Predictive Maintenance
- Edge and Embedded Artificial Intelligence
- Integration of AI into Operating Systems and System Software
- Adaptive and Self-Optimizing Software Systems

EDUCATION

Bachelor's Degree in Computer Engineering
Polytechnic of Turin

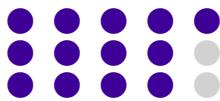
October 2013 - March 2017 Turin, TO Italy

Master's Degree in Computer Engineering
- Software and Digital Systems
Polytechnic of Turin

March 2017 - June 2020 Turin, TO Italy

LANGUAGES

Italian
English
German



PROGRAMMING LANGUAGES

- C / Embedded C / POSIX
- Modern C++ (C++20)
- C# / .NET
- Python (Scientific, ML, Backend)
- Assembly (x86, ARM)
- SQL
- Bash / Unix Tooling
- Java / Kotlin (Android)
- JavaScript (Node.js / React)
- PHP / HTML / CSS

SYSTEM ENGINEERING

- Distributed Systems
- Real-Time Systems
- Embedded Computing
- Edge Computing
- Resilient Communication

COMPETENCIES

- Systems Thinking
- Analytical Mindset
- Research-Oriented
- Architecture Analysis
- Trade-Off Analysis
- Experimental Design
- Validation Techniques
- Reliability Engineering
- Quality Engineering
- Technical Leadership
- Mentorship
- Structured Communication
- End-to-End Lifecycle Ownership

NETWORKING

- TCP/IP
- UDP
- Linux Networking Stack
- Wireless Networks
- Industrial Networks
- IEEE 802.11
- CAN
- SSH
- VPN
- BLE Protocols
- IoT Protocols
- MQTT / AMQP
- Message-Oriented Middleware
- Secure Communication
- TLS / PKI
- Network Diagnostics
- Network Analysis

FRAMEWORKS & TOOLS

- NumPy
- SciPy
- Pandas
- Flask
- Django
- BeautifulSoup
- Scikit-Learn
- Keras
- Statsmodels
- Matplotlib
- Plotly
- Seaborn
- Paho-MQTT
- ZeroMQ
- PyMongo
- MongoC
- SQLAlchemy
- TKinter
- Jupyter
- Anaconda
- WPF
- Docker
- CI/CD Pipelines
- Linux Toolchain
- GCC / Clang / CMake
- VS Code
- Visual Studio / MVSC
- PyCharm
- Eclipse
- CodeBlocks
- DevC++
- MicroKeil

Operating Systems Concepts

Linux / Kernel / POSIX

Git

GitHub

GitLab

BitBucket

Doxygen

DrawIO

Concurrency & Synchronization

Memory Management

PlantUML

LaTeX

Scientific Documentation

Performance Optimization

Cross-Compilation

ARM / AArch64 / x86

Secure System Design

Virtualization

Emulation (QEMU)

Build Systems

Dependency Management

ML & DATA SCIENCE

Anomaly Detection

Novelty Detection

Predictive Maintenance

Time Series Analysis

Statistical Modeling

Inference Techniques

Feature Engineering

Dimensionality Reduction

Supervised Learning

Unsupervised Learning

Model Validation

Performance Evaluation

Optimization Techniques

Real-Time Data Pipelines

Distributed Databases

MongoDB

ETL Processes

Data Engineering

Relational Databases

MySQL / SQL Server / SQLite

NoSQL Database

Distributed Data Architectures

Data Modeling

Query Optimization

Replication

Backup

Fault Tolerance

WORK EXPERIENCE

IT Consultant (Self-Employed)

Freelance

📅 Jan 2024 – Present

📍 Italy

- Deliver end-to-end software solutions using Python, SQL, and web frameworks (Flask, Django).
- Provide academic consulting for undergraduates on system programming, embedded systems, and AI/ML.
- Developed Arduino and x86/ARM projects demonstrating low-level hardware-software integration.

Research Fellow – Health & Wearable Software Engineer

Polytechnic of Turin • SAT / Juventus

📅 Dec 2022 – Dec 2023

📍 Turin

- Conducted R&D on wearable systems and digital health applications, focusing on sleep analysis and prediction.
- Developed Garmin Watch Applications (GWAs) leveraging sensors (HR, HRV, SpO₂, respiration, accelerometer) and FIT protocol.
- Designed and prototyped a distributed sleep prediction system integrating Garmin Watch and Raspberry Pi via BLE with GATT-based data exchange.
- Implemented a multi-domain MongoDB database and MQTT-based ingestion server for real-time biomedical data storage.
- Developed real-time visualization tools and secure analytics web services (HTTPS) for aggregated biomedical statistics.
- Maintained and enhanced Android applications integrating GWAs, BLE, ZeroMQ, and MQTT for biomedical data acquisition and streaming.

Research Fellow – Predictive Maintenance Software Engineer

Polytechnic of Turin • SPEA S.p.A.

📅 Dec 2020 – Dec 2022

📍 Turin – Volpiano

- Developed predictive maintenance solutions for Industry 4.0 Automated Test Equipment (ATE) using Python, MongoDB, and TCP/IP-based real-time data pipelines.
- Implemented multi-channel data fusion, simulation systems, and fault-tolerant databases supporting large-scale sensor and DUT datasets.
- Designed and deployed AI/ML novelty detection frameworks for early fault detection and predictive analysis.
- Co-authored publication: “A Real-Time Novelty Recognition Framework based on Machine Learning for Fault Detection”, MDPI Algorithms.
- Achieved improved data acquisition latency and real-time anomaly detection for industrial ATE systems.

Graduate Student – Master Thesis – IEEE 802.11 WiFi Industrial Networks

Implementation and Optimization of Application-Level WiFi Redundancy Techniques

📅 April 2019 – June 2020

📍 Turin, Italy

- Designed and implemented application-level WiFi redundancy techniques for IEEE 802.11 industrial wireless networks in smart manufacturing contexts.
- Developed user-space software for message duplication, retransmission handling, and reliability management.
- Implemented Reactive Duplication Avoidance (RDA) to optimize network traffic and packet queuing.

- Conducted simulated and real-world experiments, performing systematic data analysis and performance validation.

Private Teacher – Computer Science and Engineering

Freelance

📅 January 2020 – August 2020

📍 Turin, Italy

- Delivered structured theoretical and practical lessons to university and high school students in computer science and engineering.
- Covered programming languages (C, C++, Java, Python, C#), web development (PHP, HTML, JavaScript, CSS), cybersecurity, operating systems (Linux, Windows), Bash/Batch scripting, computer architecture, machine learning, AI, and database systems.
- Designed exercises and projects to reinforce practical skills in software development, system design, and problem-solving.

Trading System Developer – Automated Expert Advisor

Freelance

📅 August 2017 – September 2017

📍 Turin, Italy

- Developed an automated trading system for the MetaTrader 4 (MT4) platform using MetaQuotes Language (MQL).
- Implemented event-driven routines for initialization, deinitialization, and timer-based execution.
- Designed autonomous decision-making algorithms based on client-defined trading strategies.
- Integrated logging, reporting, and seamless API interaction for real-time trade execution and monitoring.

PUBLICATIONS

- 'A Real-Time Novelty Recognition Framework based on Machine Learning for Fault Detection', U. Albertin, G. Pedone, G. Squillero, M. Chiaberge, M. Brossa, MDPI Algorithms on call for paper 'Machine Learning and Deep Learning Applications for Anomaly and Fault Detection', November 2022.

BEYOND MY CAREER

During my leisure, I undertake personal and collaborative projects spanning different domains (e.g. ML and AI, Micro Programming and OS, Computer Networks and Database, Cyber Security and Information Systems) in order to refine my project management skills, apply and fortify my knowledge. I continuously learn about new technologies, by performing training activities, in order to stay abreast of evolving technological advancements and enrich my skills. Furthermore I design standards and develop templates for the most common documentation (technological evaluation, project proposal, design notes, product documentation, manuals) in order to maximize my productivity.

LIFE EXPERIENCES

Restaurant Waiter

Ristorante Pizzeria La Giummara

📅 June 2009 - September 2012

📍 Salemi, Italy TP

Worked as a waiter, handling various tasks including customer service, event organization, inventory management, food preparation, and team collaboration.

Lived in Germany

📅 June 1994 - April 2005

📍 Esslingen, Stuttgart

Received primary and high school education in Germany, with a focus on practical skills, multiculturalism, and self-improvement. Developed proficiency in German language and gained exposure to diverse cultures and educational experiences.